Project Title

Shopping Cart Abandonment Prediction Using Decision Trees and Random Forest

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Shopping Cart Abandonment Prediction system is a machine learning-based application designed to identify customers who are likely to leave an e-commerce website without completing their purchase. The system analyzes customer behavior using five features: browsing time, products viewed, cart value, previous purchases, and discount usage. Based on these inputs, it classifies each customer as either Cart Abandoned or Purchase Completed. Two supervised classification algorithms, Decision Tree and Random Forest, are implemented using scikit-learn and evaluated with accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and stratified 5-fold cross-validation. On a held-out test set of 600 customers from a dataset of 3,000 records, the Decision Tree achieved 74.0% accuracy and the Random Forest 73.3%, while the Random Forest showed better ranking ability with a ROC-AUC of 0.833 against 0.808. Feature importance analysis shows that cart value is the most influential factor, followed by browsing time, while discount usage contributes the least. Both models are deployed in a Flask web application that returns the abandonment probability, the decision rules followed by the tree, and risk-based advice such as cart reminders or personalised discounts. The project demonstrates the practical use of tree-based data structures, ensemble learning, and model evaluation in a real-world e-commerce problem. The proposed system can be extended with larger real-world datasets, additional behavior features, real-time prediction, and integration with e-commerce platforms.

Signature of the Guide